

Rapalogs in cancer prevention

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Common cancer is an age-related disease. Slow aging is associated with reduced and delayed carcinogenesis. Calorie restriction (CR), the most studied anti-aging intervention, prevents cancer by slowing down the aging process. Evidence is emerging that CR decelerates aging by deactivating MTOR (Target of Rapamycin). Rapamycin and other rapalogs suppress cellular senescence, slow down aging and postpone age-related diseases including cancer. At the same time, rapalogs are approved for certain cancer treatments. Can cancer prevention be explained by direct targeting of cancer cells? Or does rapamycin prevent cancer indirectly through slowing down the aging process? Increasing evidence points to the latter scenario.